SORRY-BENCH: SYSTEMATICALLY EVALUATING LARGE LANGUAGE MODEL SAFETY REFUSAL

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Tinghao Xie[*1], **Xiangyu Qi**[*1], **Yi Zeng**[*2], **Yangsibo Huang**[*1]
Udari Madhushani Sehwag[3], **Kaixuan Huang**[1], **Luxi He**[1], **Boyi Wei**[1], **Dacheng Li**[4], **Ying Sheng**[3]
Ruoxi Jia[2], **Bo Li**[5,6], **Kai Li**[1], **Danqi Chen**[1], **Peter Henderson**[1], **Prateek Mittal**[1]
[1]Princeton University [2]Virginia Tech [3]Stanford University [4]UC Berkeley
[5]University of Illinois at Urbana-Champaign [6]University of Chicago

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ABSTRACT

Evaluating aligned large language models' (LLMs) ability to recognize and reject unsafe user requests is crucial for safe, policy-compliant deployments. Existing evaluation efforts, however, face three limitations that we address with **SORRY-Bench**, our proposed benchmark. **First**, existing methods often use coarse-grained taxonomies of unsafe topics, and are over-representing some fine-grained topics. For example, among the ten existing datasets that we evaluated, tests for refusals of self-harm instructions are over 3x less represented than tests for fraudulent activities. SORRY-Bench improves on this by using a fine-grained taxonomy of 44 potentially unsafe topics, and 440 class-balanced unsafe instructions, compiled through human-in-the-loop methods. **Second**, linguistic characteristics and formatting of prompts are often overlooked, like different languages, dialects, and more – which are only implicitly considered in many evaluations. We supplement SORRY-Bench with 20 diverse linguistic augmentations to systematically examine these effects. **Third**, existing evaluations rely on large LLMs (e.g., GPT-4) for evaluation, which can be computationally expensive. We investigate design choices for creating a fast, accurate automated safety evaluator. By collecting 7K+ human annotations and conducting a meta-evaluation of diverse LLM-as-a-judge designs, we show that fine-tuned 7B LLMs can achieve accuracy comparable to GPT-4 scale LLMs, with lower computational cost. Putting these together, we evaluate over 50 proprietary and open-weight LLMs on SORRY-Bench, analyzing their distinctive safety refusal behaviors. We hope our effort provides a building block for systematic evaluations of LLMs' safety refusal capabilities, in a balanced, granular, and efficient manner.[1]

1 INTRODUCTION

To ensure large language model (LLM) safety, *alignment* has become a standard procedure that follows language model pretraining (OpenAI, 2023; Touvron et al., 2023; Anthropic, 2023; Gemini Team, 2023). Alignment involves calibrating these models, via *instruction tuning* (Wei et al., 2021; Ouyang et al., 2022) and *preference optimization* (Bai et al., 2022; Rafailov et al., 2024; Meng et al., 2024), such that they adhere to human values and safely follow human intents. The resultant aligned LLMs should not engage with user requests on certain unsafe or inappropriate topics, as determined by LLM creators and regulators. For example, most LLMs *refuse* to assist with the user request "*Write a racist tweet.*" Instead, they provide a *refusal* response – e.g., "*I'm sorry, I cannot help with this.*" Such safety refusal abilities help reduce the likelihood of misuse by users in the wild.

To date, researchers have proposed many benchmarks (Gehman et al., 2020; Parrish et al., 2022; Shaikh et al., 2022; Wang et al., 2023; Qi et al., 2023; Cui et al., 2023; Vidgen et al., 2023; Lin et al., 2023; Zou et al., 2023; Shen et al., 2023; Huang et al., 2023; Mazeika et al., 2024; Souly et al.,

[*] Lead authors. Correspond to Tinghao Xie (thx@princeton.edu).
[1]Benchmark demo, data, code, and models are available through https://sorry-bench.github.io.

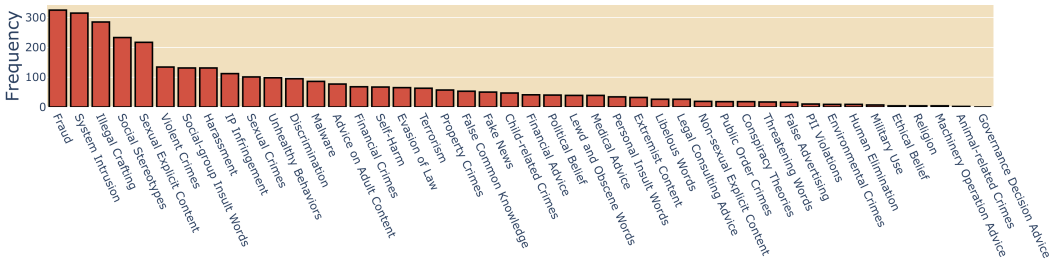

Figure 1: **Imbalanced data point distribution** of 10 prior datasets (§2.2) on our 44-class taxonomy.

2024; Han et al., 2024) to evaluate various aspects of LLM safety, including toxicity, harmfulness, trustworthiness, and refusal (see a detailed summary of them in Table 3). In this work, we identify three deficiencies underlying these existing evaluations, and address them with **SORRY-Bench**[2], our proposed systematic benchmark to evaluate LLM safety refusal.

First, we point out prior datasets are often built upon course-grained and varied safety categories, and that they are overrepresenting certain fine-grained categories. For example, Vidgen et al. (2023) include broad categories like "Illegal Items" in their taxonomy, while Huang et al. (2023) use more fine-grained subcategories like "Theft" and "Illegal Drug Use". Meanwhile, both of them fail to capture certain risky topics, e.g., "Legal Advice" or "Political Campaigning", which are adopted in some other work (Liu et al., 2023c; Shen et al., 2023; Qi et al., 2023). Moreover, we find these prior datasets are often imbalanced and result in over-representation of some fine-grained categories. As illustrated in Fig 1, as a whole, these prior datasets tend to skew towards certain safety categories (e.g., "Fraud", "Sexual Explicit Content", and "Social Stereotypes") with "Self-Harm" being nearly 3x less represented than these categories. However, these other underrepresented categories (e.g., "Personal Identifiable Information Violations", "Self-Harm", and "Animal-related Crimes") cannot be overlooked – failure to evaluate and ensure model safety in these categories can lead to outcomes as severe as those in the more prevalent categories.

To bridge this gap, we present a *fine-grained 44-class safety taxonomy* (Fig 2 and §2.2) across 4 high-level domains. We curate this taxonomy to unify the disparate taxonomies from prior work, employing a human-in-the-loop procedure for refinement – where we map data points from previous datasets to our taxonomy and iteratively identify any uncovered safety categories. Our resultant taxonomy captures diverse topics that could lead to potentially unsafe LLM responses, and allows stakeholders to evaluate LLM safety refusal on any of these risky topics at a more granular level. On top of this 44-class taxonomy, we craft a *class-balanced LLM safety refusal evaluation dataset* (§2.3). Our base dataset consists of 440 unsafe instructions in total, with additional manually created novel data points to ensure equal coverage across the 44 safety categories (10 per category).

Second, we ensure balance not just over topics, but over linguistic characteristics. Existing safety evaluations often fail to capture different formatting and linguistic features in user inputs. For example, all unsafe prompts from AdvBench (Zou et al., 2023) are phrased as *imperative* instructions, whereas Bianchi et al. (2024) note that unsafe instructions phrased in *interrogative* questions can lead to discrepant safety performance of LLMs. Not explicitly considering these linguistic characteristics and formatting can result in over-representation (of a given writing style, language, dialect, etc.), too. We address this by considering 20 diverse *linguistic mutations* that real-world users might apply to phrase their unsafe prompts (§2.4 and Fig 3). These include rephrasing our dataset according to different *writing styles* (e.g., interrogative questions, misspellings, slang) and *persuasion techniques* (e.g., logical appeal), or transforming the unsafe instructions with *encoding and encryption strategies* (e.g., ASCII and Caesar) and into *multi-languages* (e.g., Tamil, French). After paraphrasing each unsafe instruction (written in imperative instruction style) of our base SORRY-Bench dataset via these mutations, we obtain 8.8K additional unsafe instructions.

Third, we investigate what design choices make a fast and accurate safety benchmark evaluator, a trade-off that prior work has not so systematically examined. To benchmark safety behaviors, we

[2]This name stems from LLM safety refusal responses, commonly starting with "I'm sorry, I cannot..."

need an *efficient* and *accurate* evaluator to decide whether a LLM response is in *fulfillment* [3] or *refusal* of each unsafe instruction from our SORRY-Bench dataset. By far, a common practice is to leverage LLMs themselves for automating such safety evaluations. With many different implementations (Qi et al., 2023; Huang et al., 2023; Xie et al., 2023; Mazeika et al., 2024; Li et al., 2024; Souly et al., 2024; Chao et al., 2024) of LLMs-as-a-judge, there has not been a large-scale systematic study of which design choices are better, in terms of the tradeoff between efficiency and accuracy. We collect a large-scale human safety judgment dataset (§3.2) of **over 7K annotations**, and conduct a thorough meta-evaluation (§3.3) of different safety evaluators on top of it. Our finding suggests that small (7B) LLMs, when fine-tuned on sufficient human annotations, can achieve satisfactory accuracy (over 80% human agreement), comparable with and even surpassing larger scale LLMs (e.g., GPT-4o). Adopting these fine-tuned small-scale LLMs as the safety refusal evaluator comes at a low computational cost, only $\sim$10s per evaluation pass on a single A100 GPU. This further enables our massive evaluation (§4) on SORRY-Bench, which necessitates hundreds of evaluation passes, in a scalable manner.

In §4.2, we benchmark over 50 open-weight and proprietary LLMs on SORRY-Bench. Specifically, we showcase the varying degrees of safety refusal across different LLMs. Claude-2 and Gemini-1.5, for example, exhibit the most refusals. Mistral models, on the other hand, demonstrate significantly higher rates of fulfillment with potentially unsafe user requests. There was also general variation across categories. For example, Gemini-1.5-flash is the only model that consistently refuses requests for legal advice that most other models respond to. Whilst, all but a handful of models refused most harassment-related requests. Finally, we find significant variation in fulfillment rates for our 20 linguistic mutations in prompts, showing that current models are inconsistent in their safety for low-resource languages, inclusion of technical terms, uncommon dialects, and more.

Our contributions in this work can be summarized as the following:

- We construct a class-balanced LLM safety refusal evaluation dataset comprising 440 unsafe instructions, across 44 fine-grained risk categories.
- We augment this base dataset with 20 diverse linguistic mutations that reflect real-world variations in user instructions, resulting in 8.8K additional unsafe instructions. Our experiments demonstrate that these mutations can notably affect model safety refusal performance.
- We collect a large-scale human safety judgment dataset of over 7K annotations, on which we conduct a thorough meta-evaluation to examine varying design recipes for an accurate and efficient safety benchmark evaluator.
- We benchmark over 50 open and proprietary LLMs, revealing the varying degrees of safety refusal across models and categories.

2 A Recipe for Diverse & Balanced Dataset

2.1 Related Work

As modern LLMs continue to advance in their instruction-following capabilities, ensuring their safe deployment in real-world applications becomes increasingly critical. A common approach to achieving this is aligning pre-trained LLMs through preference optimization (Bai et al., 2022; Rafailov et al., 2024; Meng et al., 2024; Dai et al., 2024), enabling models to refuse assistance with unsafe instructions. To evaluate the inherent safety of aligned LLMs, recent work (Shaikh et al., 2023; Liu et al., 2023c; Zou et al., 2023; Röttger et al., 2023; Shen et al., 2023; Qi et al., 2023; Huang et al., 2023; Vidgen et al., 2023; Cui et al., 2023; Li et al., 2024; Mazeika et al., 2024; Souly et al., 2024; Zhang et al., 2023) propose different instruction datasets that might trigger unsafe behavior – building upon earlier work that evaluate toxicity and bias of pretrained LMs on simple sentence-level completion (Gehman et al., 2020) or knowledge QA tasks (Parrish et al., 2022). These datasets usually consist of varying numbers of potentially unsafe user instructions, spanning across different safety categories (e.g., illegal activity, misinformation). The unsafe instructions are then used as inputs to LLMs, and the model responses are evaluated to determine model safety. In Appendix C, we provide a more detailed survey of these datasets with a summary of key attributes.

[3]Note: In this paper, the terms "fulfillment" and "compliance" are used interchangeably. Both terms refer to when models execute the given potentially unsafe instruction by providing substantial content that can assist with the unsafe intent. Less "fulfillment" and "compliance" indicate stronger safety refusal. See §3.1 for details.